

Colon & Colorectal Cancer — Biomarker Testing Intelligence

Alert Window: Last 30 Days · Generated: 2026-05-29 07:30:01

Situation Summary

Stage IV: 9 of 40
Biomarker Count: 22 entries
Top Targets: ctDNA (9) · MSI/MMR (5) · BRAF (3)
Breakthrough: 13 entries scored ≥ 4

Research & Guidelines Timeline

Biomarker testing entries ranked by significance score. Higher scores indicate greater clinical relevance. Always discuss with your oncologist.

HIGH

Stage IV

ctDNA

VERIFIED

LIQUID BIOPSY

PUBLISHED MAY 26, 2026

9

Circulating tumor DNA-based assessment of disease progression after liver resection or transplantation for metastatic colorectal cancer or hepatocellular carcinoma-study protocol for a randomized, controlled trial

Circulating tumor DNA (ctDNA) monitoring is being evaluated for its clinical utility in detecting minimal residual disease (MRD) after liver resection or transplantation for metastatic colorectal cancer (CRLM) and hepatocellular carcinoma (HCC). This randomized controlled trial aims to determine if ctDNA can provide earlier detection of disease progression compared to conventional imaging and biomarkers, which have limited sensitivity.

pubmed.ncbi.nlm.nih.gov

HIGH

All Stages

ctDNA

VERIFIED

LIQUID BIOPSY

PUBLISHED MAY 18, 2026

9

Circulating Tumor DNA and Precision Biomarkers in Colorectal Cancer: Implications for Diagnosis, Monitoring, and Management of Advanced Disease

Circulating tumor DNA (ctDNA) and novel precision biomarkers enhance the diagnosis, monitoring, and management of advanced colorectal cancer (CRC). Traditional markers like CEA, KRAS/NRAS, BRAF, and MSI status are inadequate for comprehensive risk stratification. The study highlights the potential of emerging biomarkers from liquid biopsies and stool-based methylation to improve individualized treatment strategies in CRC.

pubmed.ncbi.nlm.nih.gov

HIGH

All Stages

ctDNA

VERIFIED

LIQUID BIOPSY

PUBLISHED MAY 13, 2026

9

Liquid Biopsy in Colorectal Cancer: Future Perspectives Through the Lens of Artificial Intelligence-A Comprehensive Review of Novel Literature

Liquid biopsy, analyzing circulating tumor DNA (ctDNA), circulating tumor cells (CTCs), and exosomes, offers a non-invasive alternative to traditional tissue biopsy for colorectal cancer (CRC) tumor profiling. The review highlights the limitations of tissue biopsy, such as tumor heterogeneity and invasiveness, and discusses the potential of liquid biopsy in improving CRC diagnosis and monitoring. The integration of artificial intelligence in this context may enhance the utility of liquid biopsy in clinical practice.

HIGH

All Stages

MSI/MMR

VERIFIED

LIQUID BIOPSY

PUBLISHED MAY 27, 2026

7

Evaluation of Droplet Digital PCR Assay for the Detection of Microsatellite Instability in Colorectal, Gastric, and Endometrial Cancers

Droplet digital PCR (ddPCR) was evaluated for detecting microsatellite instability (MSI) using BAT-26, ACVR2A, and DEFB105A/B markers in colorectal, gastric, and endometrial cancers. The study demonstrated that ddPCR is a highly sensitive method for MSI detection, which is crucial for diagnosing Lynch syndrome and informing immunotherapy decisions.

pubmed.ncbi.nlm.nih.gov

HIGH

All Stages

ctDNA

VERIFIED

LIQUID BIOPSY

PUBLISHED MAY 24, 2026

7

Practical integration of ctDNA-defined minimal residual disease in gastrointestinal cancers: testing windows and evidence-aligned management frameworks

Circulating tumor DNA (ctDNA) serves as a biomarker for minimal residual disease and tumor burden in gastrointestinal cancers, including colorectal cancer. This narrative review discusses practical testing windows, assay selection, and evidence-based management frameworks for integrating ctDNA into clinical practice. The review highlights that the strongest evidence supports ctDNA utilization in resected gastrointestinal cancers, indicating its potential role in monitoring and guiding treatment decisions.

pubmed.ncbi.nlm.nih.gov

HIGH

All Stages

ctDNA

VERIFIED

LIQUID BIOPSY

PUBLISHED MAY 15, 2026

6

A novel liquid-liquid cfDNA extraction method for targeted sequencing with colorectal cancer patient samples: a pilot study

A novel liquid-liquid extraction method for circulating tumor DNA (ctDNA) from plasma was developed to enhance the quality and quantity of cell-free DNA for targeted sequencing in colorectal cancer. This pilot study addresses challenges in ctDNA mutational analysis, potentially improving non-invasive detection and monitoring of gene mutations associated with colorectal cancer.

pubmed.ncbi.nlm.nih.gov

MEDIUM

Stage IV

BROAD SCOPE

VERIFIED

RESEARCH

PUBLISHED MAY 13, 2026

4

Late-line Systemic Therapy in Metastatic Colorectal Cancer: A Narrative Review and Australian Expert Perspective

Molecular profiling in metastatic colorectal cancer (mCRC) facilitates personalized treatment selection, particularly in late-line therapy. Targeted therapies, including BRAF inhibitors, anti-HER2 agents, and KRAS^{G12C} inhibitors, are effective for patients with specific molecular alterations. This narrative review highlights the importance of these advancements in expanding treatment options for mCRC patients experiencing disease progression after multiple therapies.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

HER2

VERIFIED

GENOMIC PROFILING

PUBLISHED MAY 12, 2026

4

Comprehensive genomic landscape of ERBB2 in Chinese GI tumors: mutation-centered landscapes and precision treatment opportunities

ERBB2 alterations in colorectal cancer (CRC) and gastric cancer (GC) predominantly occur in kinase domain hotspots, influenced by tumor-specific genomic contexts. Oncogenic ERBB2 mutations correlate with higher tumor mutational burden (TMB) and microsatellite instability-high (MSI-H) status. These findings support the clinical evaluation of mutation-selective HER2 inhibitors and their potential combination with immune checkpoint blockade for targeted therapy in CRC and GC.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

BROAD SCOPE

VERIFIED

RESEARCH

PUBLISHED MAY 27, 2026

3

Immune microenvironment evolution across the serrated neoplasia pathway and its relevance to immunotherapy

Serrated neoplasia pathway tumors, comprising 15%-30% of sporadic colorectal cancers, exhibit early BRAF V600E mutations and distinct immune microenvironments. Despite the efficacy of immune checkpoint inhibitors in MSI-H/dMMR colorectal cancer, these tumors demonstrate significant molecular heterogeneity, impacting their response to immunotherapy. Understanding the immune microenvironment evolution in this context is crucial for optimizing treatment strategies.

pubmed.ncbi.nlm.nih.gov

MEDIUM

Stage IV

BRAF

VERIFIED

RESEARCH

PUBLISHED MAY 20, 2026

⚡ 3

KRAS and BRAF^{V600E} mutations modify adjuvant chemotherapy outcomes in early stage colorectal cancer

KRAS and BRAF^{V600E} mutations were assessed for their impact on adjuvant chemotherapy outcomes in stage III and high-risk stage II colorectal cancer. The study analyzed patients post-curative resection with molecular profiling. Results indicate that these mutations may modify the survival benefit associated with specific adjuvant chemotherapy regimens, suggesting a need for tailored treatment approaches based on mutation status in early-stage CRC.

pubmed.ncbi.nlm.nih.gov

HIGH

Stage I

BROAD SCOPE

VERIFIED

GUIDELINE

PUBLISHED MAY 04, 2026

⚡ 3

Truncating Adenomatous Polyposis Coli Mutations Serve as a Biomarker for Advanced Tumor Stage in Iranian Patients with Familial Colorectal Cancer

Truncating mutations in the Adenomatous Polyposis Coli (APC) gene's Mutation Cluster Region (MCR) were identified as biomarkers for advanced tumor stage in a cohort of 96 Iranian patients with familial colorectal cancer. The study utilized sequencing to classify variants according to ACMG guidelines, highlighting the prognostic significance of specific APC mutations in this population.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

BROAD SCOPE

VERIFIED

RESEARCH

PUBLISHED MAY 22, 2026

⚡ 2

Assessment of colorectal cancer recurrence risk following solid organ transplantation

Solid organ transplant (SOT) recipients exhibit a trend towards increased colorectal cancer (CRC) recurrence risk compared to population benchmarks. KRAS, APC, and BRAF mutational statuses were found to influence recurrence risk. While no direct association was observed between immunosuppression variables and recurrence, diminished immune surveillance may contribute to this elevated risk. Strategies such as reducing immunosuppression in post-SOT CRC patients could potentially mitigate recurrence.

pubmed.ncbi.nlm.nih.gov

HIGH

All Stages

BROAD SCOPE

VERIFIED

GUIDELINE

PUBLISHED MAY 18, 2026

⚡ 2

Rectifying referrals: genetics testing is underutilized in rectal cancer patient care

MSI/MMR testing recommendations are not consistently followed at the facility level, and patient compliance with genetic referrals in rectal cancer is low. When recommendations are adhered to, patients with germline mutations experience beneficial adjustments in disease management.

pubmed.ncbi.nlm.nih.gov

MEDIUM

Stage III

BRAF

VERIFIED

RESEARCH

PUBLISHED MAY 11, 2026

⚡ 2

Case Report: Cerebral metastasis following standard therapy in BRAF/KRAS wild-type colorectal cancer: an unusual presentation of disease progression

Cerebral metastasis can occur in colorectal cancer (CRC) with wild-type KRAS/BRAF, despite its association with low risk for central nervous system dissemination. A case of a 66-year-old woman with sigmoid colon adenocarcinoma illustrates this unusual progression following standard therapy. The patient initially presented with locoregionally advanced disease and distant metastasis, highlighting the need for vigilance in monitoring CNS involvement even in cases with typically favorable molecular features.

MEDIUM

All Stages

HER2

VERIFIED

GENOMIC PROFILING

PUBLISHED APR 30, 2026

⚡ 2

Individualised prediction of HER2 status in colorectal cancer: development and validation of a radiomics prediction model using (18)F-fluorodeoxyglucose positron emission tomography/magnetic resonance ((18)F-FDG PET/MR)

PET/MR-based radiomics provides a noninvasive method for preoperative evaluation of HER2 status in colorectal cancer. The study developed and validated a prediction model, indicating its potential utility in clinical decision-making regarding targeted therapies.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

RAS

VERIFIED

RESEARCH

PUBLISHED MAY 25, 2026

⚡ 1

RAS mutation status and immune microenvironment define distinct prognostic landscapes and predict chemotherapy benefit in pMMR colorectal cancer

RAS mutational status influences the immune microenvironment in pMMR colorectal cancer, affecting prognosis and chemotherapy response. The study suggests that combining RAS status with tumor immune microenvironment features, especially CD163, may enhance patient selection for chemotherapy benefit.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

BROAD SCOPE

VERIFIED

GENOMIC PROFILING

PUBLISHED MAY 22, 2026

⚡ 1

Towards quality control and harmonization of deep learning CT radiomics: An in-silico feasibility study with virtual colorectal liver metastases

Deep learning (DL)-based radiomic biomarker prediction for colorectal liver metastases showed improved reproducibility with image harmonization and provided a reliability metric through uncertainty estimation in a controlled in-silico study. While these findings indicate methodological feasibility, clinical validation on actual CT data is necessary for translation into practice.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

KRAS

VERIFIED

RESEARCH

PUBLISHED MAY 21, 2026

⚡ 1

Frequent Detection of KRAS-G12C and PIK3CA-Q546K Mutations in MAP Tumors Highlights their Role in MUTYH Variants of Uncertain Significance Reclassification

KRAS-G12C and PIK3CA-Q546K mutations are frequently detected in colorectal cancers associated with biallelic pathogenic variants in MUTYH, contrasting with their rarity in sporadic cases. This enrichment suggests that these mutations may aid in the reclassification of MUTYH variants of uncertain significance. The study highlights the potential role of these specific biomarkers in the context of MUTYH-associated polyposis.

pubmed.ncbi.nlm.nih.gov

MEDIUM

Stage III

MSI/MMR

VERIFIED

RESEARCH

PUBLISHED MAY 15, 2026

⚡ 1

Preoperative MMR testing in colorectal cancer: Large-scale cohort confirms reliability but identifies tumor heterogeneity as a challenge

Preoperative biopsy for MMR assessment in colorectal cancer demonstrates high reliability in a large cohort. However, tumor heterogeneity, including weak staining and intraglandular variability, poses challenges for accurate interpretation. Implementing refined criteria can enhance the evaluation of atypical MMR patterns. Neoadjuvant therapy does not significantly affect the reliability of MMR interpretations, though clinicians should be aware of rare therapy-associated alterations.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

MSI/MMR

VERIFIED

RESEARCH

PUBLISHED MAY 13, 2026

⚡ 1

Case Report: Synchronous colorectal adenocarcinomas with discordant mismatch repair status: a case of lynch-like syndrome and serrated pathway association

Synchronous colorectal adenocarcinomas (SCRCs) with discordant mismatch repair (MMR) status require lesion-specific molecular characterization and regional lymph node MMR phenotyping for optimal management. This case report outlines a diagnostic workflow and surgical decision-making framework, emphasizing the need for risk-adapted individualized therapy in cases of molecularly heterogeneous colorectal cancer.

pubmed.ncbi.nlm.nih.gov

MEDIUM

All Stages

BROAD SCOPE

VERIFIED

RESEARCH

PUBLISHED MAY 10, 2026

⚡ 1

RNF43-mutations Are Associated With the Classical Molecular Subtype, Vigorous Antitumor Immune Responses and Prolonged Survival in Pancreatic Adenocarcinoma

RNF43 mutations in pancreatic ductal adenocarcinoma (PDAC) were assessed using next-generation sequencing (OncoPrint TML-Assay). The study found that RNF43 mutations are associated with the classical molecular subtype, robust antitumor immune responses, and prolonged survival. Additionally, these mutations correlated with microsatellite stability and resulted in fewer and later recurrences in PDAC.

pubmed.ncbi.nlm.nih.gov

Appendix A — Patient-Friendly Summary

Plain-language overview for patients, caregivers, and family members

What's happening

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. The results determine which FDA-approved treatments may be available to you. The entries below represent the highest-scoring study per biomarker target drawn from the full research archive — not limited to the current report window. Some entries may predate the timeline above; they are included because they remain the most clinically significant reference available for that biomarker.

KRAS

Circulating tumor DNA (ctDNA) clearance as an early indicator of sotorasib + panitumumab efficacy and prognosis in KRAS G12C–mutated colorectal cancer (mCRC): Results from phase 3 CodeBreak 300.

In the phase 3 CodeBreak 300 trial, ctDNA clearance was assessed as an early indicator of treatment efficacy and prognosis in KRAS G12C–mutated mCRC patients receiving sotorasib and panitumumab. The findings suggest that ctDNA clearance may serve as a valuable biomarker for monitoring therapeutic response and predicting outcomes in this patient population.

ctDNA

Circulating tumor DNA-based assessment of disease progression after liver resection or transplantation for metastatic colorectal cancer or hepatocellular carcinoma-study protocol for a randomized, controlled trial

Circulating tumor DNA (ctDNA) monitoring is being evaluated for its clinical utility in detecting minimal residual disease (MRD) after liver resection or transplantation for metastatic colorectal cancer (CRLM) and hepatocellular carcinoma (HCC). This randomized controlled trial aims to determine if ctDNA can provide earlier detection of disease progression compared to conventional imaging and biomarkers, which have limited sensitivity.

TP53

Clinical impact of TP53 mutation status on survival outcomes in metastatic colorectal cancer

TP53 mutations serve as an independent prognostic marker for prolonged progression-free survival in metastatic colorectal cancer. While overall survival differences were not statistically significant, the findings suggest the need for further validation in prospective studies to assess TP53's role in therapeutic stratification.

BRAF

Detection of KRAS, NRAS and BRAF Mutations in Liquid Biopsy from Patients with Colorectal Cancer

ddPCR analysis of circulating tumor DNA (ctDNA) in colorectal cancer patients reveals that it can provide complementary information for molecular diagnosis. The study highlights the utility of liquid biopsy in detecting KRAS, NRAS, and BRAF mutations, enhancing diagnostic accuracy in this patient population.

MSI/MMR

Association of progression-free survival (PFS) with microsatellite instability–high (MSI-H) or mismatch repair–deficient (dMMR) status determined by single vs dual validated companion diagnostic (CDx) assays: CheckMate-8HW post hoc analyses.

In post hoc analyses of CheckMate-8HW, progression-free survival (PFS) was assessed in relation to microsatellite instability–high (MSI-H) or mismatch repair–deficient (dMMR) status using single versus dual validated companion diagnostic (CDx) assays. The findings suggest that the method of biomarker determination may influence PFS outcomes in colorectal cancer patients.

NGS

Exploratory biomarker findings from regorafenib plus toripalimab in patients with refractory metastatic colorectal cancer (REGOTORI study)

In the REGOTORI study, a multi-omics analysis identified several biomarkers associated with regorafenib plus toripalimab in refractory metastatic colorectal cancer. Key findings include ctDNA maxAF, HAF-bTMB, b1TH, HAF-b1TH, and mutational status of SMAD4 or PIK3CA, along with TME markers, which may serve as predictive or prognostic indicators for treatment response.

HER2

Comprehensive genomic landscape of ERBB2 in Chinese GI tumors: mutation-centered landscapes and precision treatment opportunities

ERBB2 alterations in colorectal cancer (CRC) and gastric cancer (GC) predominantly occur in kinase domain hotspots, influenced by tumor-specific genomic contexts. Oncogenic ERBB2 mutations correlate with higher tumor mutational burden (TMB) and microsatellite instability-high (MSI-H) status. These findings support the clinical evaluation of mutation-selective HER2 inhibitors and their potential combination with immune checkpoint blockade for targeted therapy in CRC and GC.

RAS

Effect of sex differences on the emergence of ctDNA RAS mutations in RAS wild-type colorectal cancer

In a single-center retrospective study of 43 patients with microsatellite stable RAS and BRAF wild-type metastatic colorectal cancer, the emergence of RAS mutations during chemotherapy was analyzed using a ctDNA assay. The study aimed to identify clinicopathological factors associated with this emergence, highlighting the potential impact of sex differences on RAS mutation development in RAS wild-type mCRC.

Genomic Profiling

Integrated Molecular Profiling of Colorectal Cancer by Tumor Location: Evidence from a Real-World Cohort with Primary and Metastatic Samples

Colorectal cancer (CRC) tumor location correlates with distinct molecular profiles. Right-sided tumors exhibit dMMR, MSI-H, and elevated TMB, while left-sided tumors show increased ERBB2 alterations and class 5 APC mutations. These findings underscore the necessity of incorporating tumor location into personalized molecular diagnostics and treatment strategies for CRC patients.

EGFR

Circulating Butyrate Attenuates Cetuximab Efficacy in Colorectal Cancer Through EGFR and AMPK-Wip1 Signaling

Butyrate may serve as a predictive biomarker for treatment response in KRAS wild-type colorectal cancer, as it reduces cetuximab efficacy via EGFR and AMPK-Wip1 signaling pathways.

miRNA

Diagnostic Performance of Circulating miRNA-92a and Related Circulating miRNA Panels for Colorectal Cancer: An Updated Systematic Review and Meta-Analysis

Circulating miRNA-92a shows superior diagnostic performance as a non-invasive biomarker for early colorectal cancer detection. The systematic review and meta-analysis suggest its potential utility, though further large-scale prospective studies are needed to standardize testing protocols and confirm clinical applicability.

TMB

Plasma-TMB (pTMB) and circulating tumor fraction (ctF) dynamics as biomarkers of benefit from the addition of atezolizumab to first-line FOLFOXIRI/bevacizumab in metastatic colorectal cancer (mCRC): A translational analysis of the AtezoTRIBE study.

In the AtezoTRIBE study, plasma tumor mutational burden (pTMB) and circulating tumor fraction (ctF) were analyzed as biomarkers for the efficacy of adding atezolizumab to first-line FOLFOXIRI/bevacizumab in metastatic colorectal cancer. The study suggests that pTMB and ctF dynamics may predict treatment benefit, providing insights for personalized therapy strategies in mCRC.

IHC

Integration of deep learning and radiomic features from multiplex immunohistochemistry images for reproducible Multi-Outcome prediction in a Multi-Center study of colorectal cancer

Fused multiplex immunohistochemistry (mIHC)-derived radiomic and deep learning features provide accurate and interpretable predictions for multiple colorectal cancer outcomes. The study supports the integration of these features into precision oncology workflows, enhancing the potential for reproducible multi-outcome predictions across different centers.

Stool DNA

An algorithm-enhanced stool DNA system improves the differential diagnosis of colorectal cancer versus Crohn's disease in high-risk symptomatic patients

The FIT-sDNA-CA system, which integrates genetic, epigenetic, and inflammatory biomarkers, enhances the differential diagnosis of colorectal cancer versus Crohn's disease in high-risk symptomatic patients. This algorithm-enhanced stool DNA testing improves specificity compared to traditional fecal DNA tests, serving as an accurate non-invasive triage tool that aids in early risk stratification and reduces unnecessary endoscopic referrals.

What this means for you

If you have been diagnosed with colorectal cancer, ask your oncologist: *"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"* This is especially important if you have been diagnosed with metastatic (Stage IV) colorectal cancer, where biomarker results directly determine which targeted therapies and immunotherapies are available to you. Based on the research in this report, the most actively studied biomarkers right now are **KRAS, ctDNA, TP53, BRAF, and MSI/MMR**. Testing should happen at diagnosis — before treatment begins — because results take 1–2 weeks and can shape your first treatment decision.

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This is not medical advice. Always speak with your oncologist before making any treatment decisions.

Appendix B — Colon & Colorectal Cancer: Biomarker Testing Reference

What Is Biomarker Testing?

Biomarker testing — also called molecular profiling or genomic testing — analyzes your tumor's DNA to identify specific mutations. A single next-generation sequencing (NGS) panel can check many gene mutations at once. Results typically take 1–2 weeks and should be obtained at diagnosis, before treatment begins.

Key Biomarkers — Test All mCRC at Diagnosis

RAS (KRAS/NRAS) · BRAF V600E · MSI/MMR status · HER2 · NTRK fusion · TMB

Testing Methods

NGS (next-generation sequencing) — checks many mutations at once from tumor tissue

Liquid biopsy / ctDNA — blood test that detects tumor DNA; useful when tissue is unavailable

IHC (immunohistochemistry) — used to test MMR status and HER2

PCR — used for specific mutation detection (e.g. MSI, KRAS)

Questions to Ask Your Oncologist

"Has my tumor been tested for all the biomarkers that have an FDA-approved treatment?"

"Was my sample taken from the primary tumor or a metastatic site — and does it need to be retested?"

"How long will results take, and will they affect my first treatment decision?"

"Is a liquid biopsy appropriate for my situation?"

Abbreviations

BRAF V600E — a specific gene mutation found in ~8–10% of colorectal cancers

ctDNA — circulating tumor DNA; tumor DNA fragments found in the bloodstream

dMMR — mismatch repair deficient; indicates the tumor may respond to immunotherapy

HER2 — a protein that drives growth in ~2–3% of colorectal cancers

IHC — immunohistochemistry; a lab staining method used to detect proteins in tissue

KRAS G12C — a specific gene mutation found in ~3–4% of colorectal cancers

mCRC — metastatic colorectal cancer (Stage IV)

MSI-H — microsatellite instability-high; indicates the tumor may respond to immunotherapy

NGS — next-generation sequencing; a method that checks many gene mutations at once

NTRK — a rare gene fusion found in <1% of colorectal cancers

RAS — a gene family (KRAS/NRAS) whose mutation status affects treatment selection

TMB — tumor mutational burden; a measure of how many mutations are in a tumor

Key Resources

ClinicalTrials.gov · NCCN Guidelines (nccn.org) · Colorectal Cancer Alliance (ccalliance.org) · Fight Colorectal Cancer (fightcrc.org) · ACS (cancer.org)

Colon & Colorectal Cancer — Biomarker Testing Intelligence

About This Report

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Limitations

This report does not represent a complete picture of all published research. Database coverage, feed availability, and relevance filtering affect what appears. Research findings may not yet reflect current clinical guidelines. Always verify information at the original source before making clinical decisions.

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https://petroai1492.github.io/delta/timelines/coloncancer_biomarker_timeline.pdf